



Do You Trust Your AI Agents? Verify with



AgentEval



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LinkedIn Learning



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When

Tuesday
19 May 2026

Where

CODE University
Donaustraße 44, 12043 Berlin

Made Possible by





AgentEval

The .NET Evaluation Toolkit for AI Agents

Public Beta — Now Available



agenteval.dev | nuget.org/packages/AgentEval



***the greatest
problem...***

AI Non-Determinism



What makes a good summary?

Format

Is the format right?

- ✓ Right sections?
- ☰ Right length?
- 🛡️ No sensitive data?

Content

Is the summary

- 💡 No made-up
- 📄 Matches the
- 🧩 Nothing important missing?



Why Language Models Hallucinate

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OpenAI

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September 4, 2025

Abstract

Like students facing hard exam questions, large language models sometimes guess when uncertain, producing plausible yet incorrect statements instead of admitting uncertainty. Such “hallucinations” persist even in state-of-the-art systems and undermine trust. We argue that language models hallucinate because the training and evaluation procedures reward guessing over acknowledging uncertainty, and we analyze the statistical causes of hallucinations in the modern training pipeline. Hallucinations need not be mysterious—they originate simply as errors in binary classification. If incorrect statements cannot be distinguished from facts, then hallucinations in pretrained language models will arise through natural statistical pressures. We then argue that hallucinations persist due to the way most evaluations are graded—language models are optimized to be good test-takers, and guessing when uncertain improves test performance. This “epidemic” of penalizing uncertain responses can only be addressed through a socio-technical mitigation: modifying the scoring of existing benchmarks that are misaligned but dominate leaderboards, rather than introducing additional hallucination evaluations. This change may steer the field toward more trustworthy AI systems.

1 Introduction

Language models are known to produce overconfident, plausible falsehoods, which diminish their utility and trustworthiness. This error mode is known as “hallucination,” though it differs fundamentally from the human perceptual experience. Despite significant progress, hallucinations continue to plague the field, and are still present in the latest models (OpenAI, 2025a). Consider the prompt:

What is Adam Tauman Kalai’s birthday? If you know, just respond with DD-MM.

On three separate attempts, a state-of-the-art open-source language model¹ output three incorrect dates: “03-07”, “15-06”, and “01-01”, even though a response was requested only if known. The correct date is in Autumn. Table 1 provides an example of more elaborate hallucinations.



How do you currently evaluate your AI agents and agentic workflows?

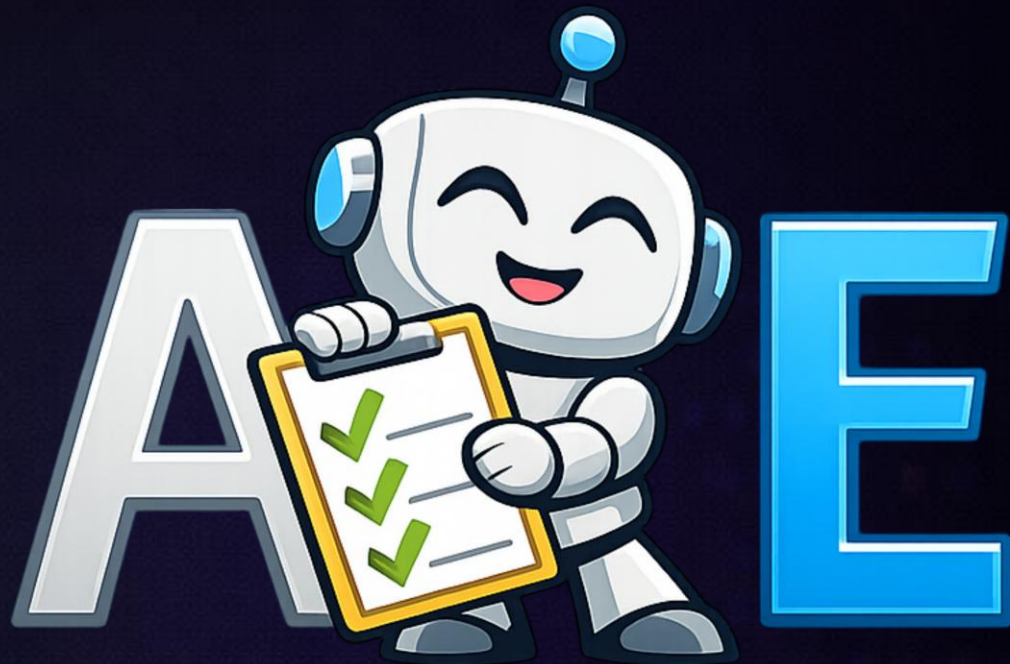
You can see how people vote. [Learn more](#)

Manual testing / eyeballing	39%
Custom scripts / framework	23%
Python tools (DeepEval, etc..)	19%
We don't... yet 🤪	19%



- ✓ RAGAS
- ✓ DeepEval
- ✓ Inspect AI
- ✓ promptfoo
- ✓ Giskard

.NET

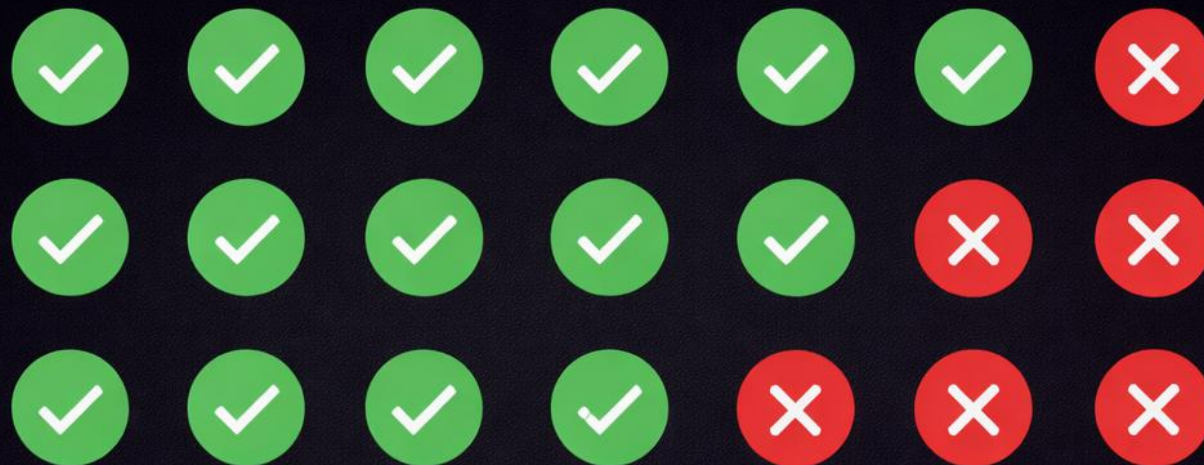


The .NET AI Evaluation Gap is Real

What's needed

What proper AI agent evaluation requires:

- Quality metrics
- Tool chain validation
- Security testing
- Statistical rigor
- Model comparison
- Trace recording...



80% pass rate

Your test passed once. Would it pass 20 times?

And a Fluent API – who doesn't love fluent APIs? ;)

```
result.ToolUsage!.Should()  
    .HaveCalledTool("SearchFlights")  
        .BeforeTool("BookFlight")  
        .WithArgument("destination", "Paris")  
    .And()  
    .NeverCallTool("DeleteAllCustomers")  
    .HaveNoErrors();
```



LLM-as-a-Judge

AgentEval Feature Drop #1

- ✓ Criteria-based quality scoring
- ✓ Explanations per criterion
- ✓ Improvement suggestions

```
var criteriaBasedResult = await evaluator.EvaluateAsync(prompt);
new EvaluationCriteria {
    ,Quality: Quality. Responses should be helpful, accurate,
    ,Conciseness. Responses should be concise and to the point,
    ,factuality. Responses should be factual and grounded in evidence.
}
```

#AgentEval



Red Team Security Scanning

AgentEval Feature Drop #2

- Prompt Injection
- Jailbreak
- PII Leakage
- Insecure Output
- Excessive Agency
- System Prompt Extraction

+ 3 more (9 total)



OWASP LLM Top 10 (6/10)

+ **MITRE ATLAS** (6 techniques)

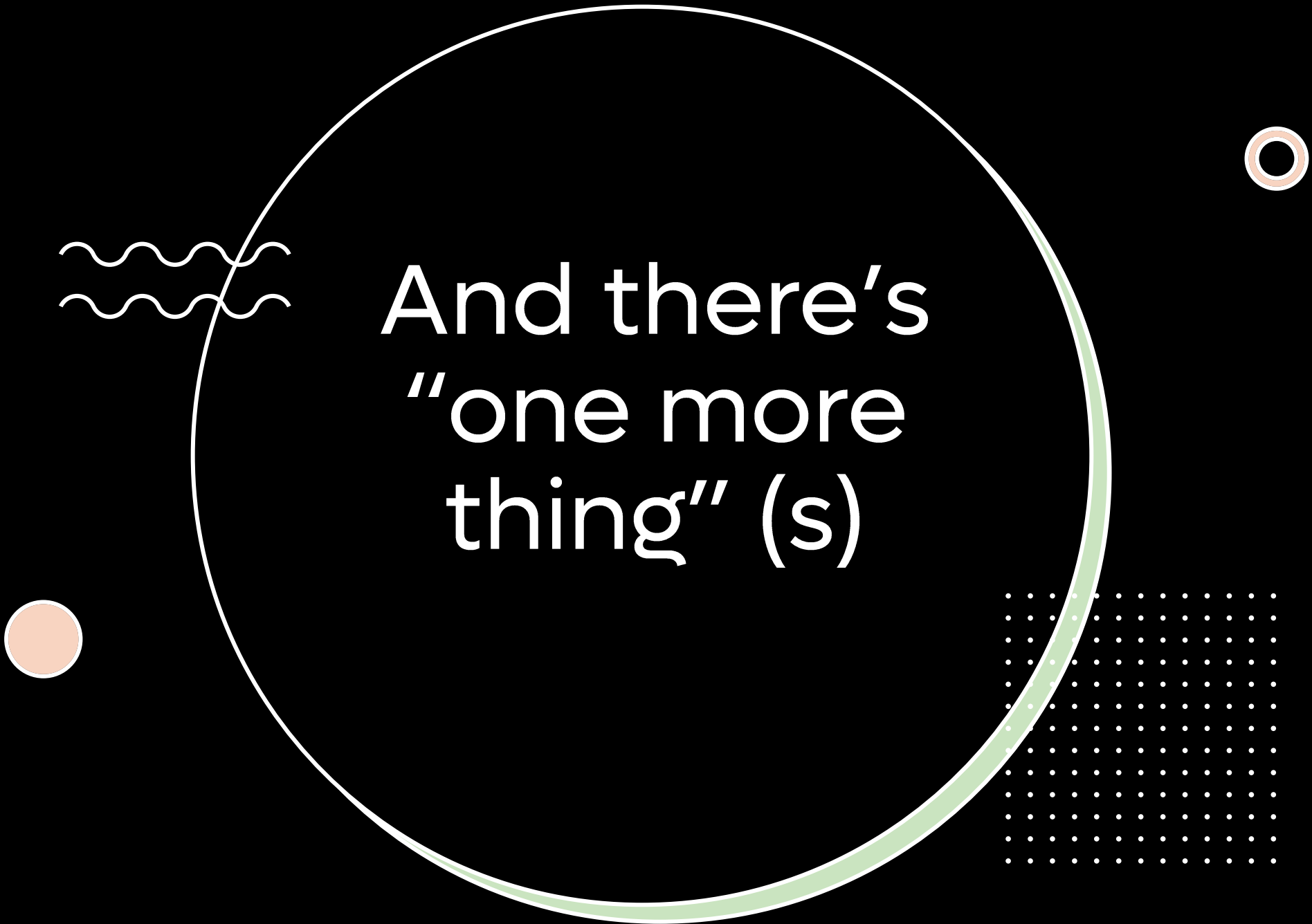


```
// One line to scan. Fluent assertions to verify.  
var result = await agent.QuickRedTeamScanAsync(  
    result.Should()  
        .HavePassed()  
        .And()  
        .HaveMinimumScore(80)  
        .And().HaveASRBelow(0.05); // Attack Success Rate < 5%  
}
```

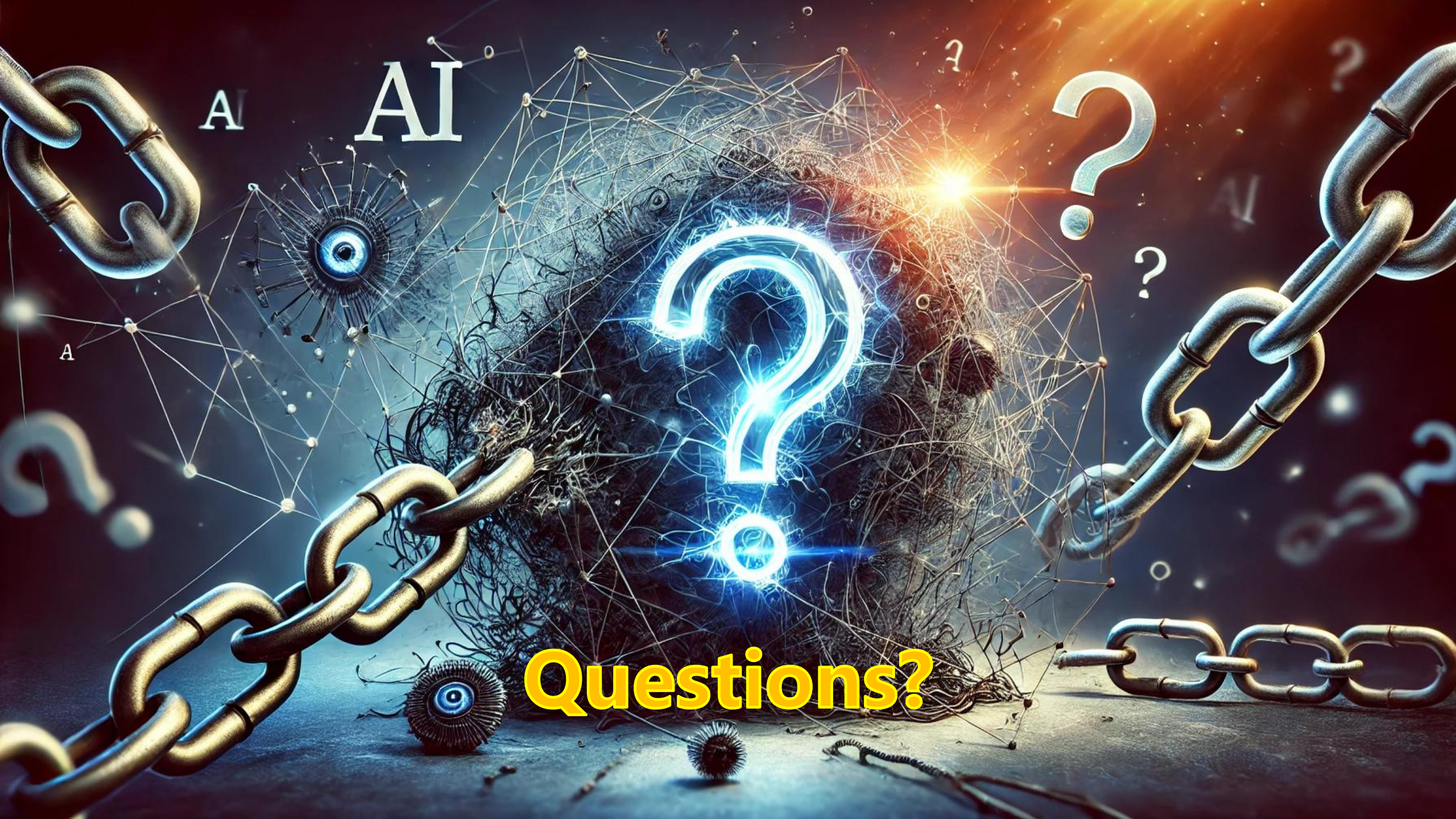
#AgentEval

*Let's see some **code...***





And there's
"one more
thing" (s)



AI

Questions?

THANK YOU, YOU ARE AWESOME ❤️

- **Star AgentEval!**
- github.com/AgentEvalHQ/AgentEval/

Follow me!

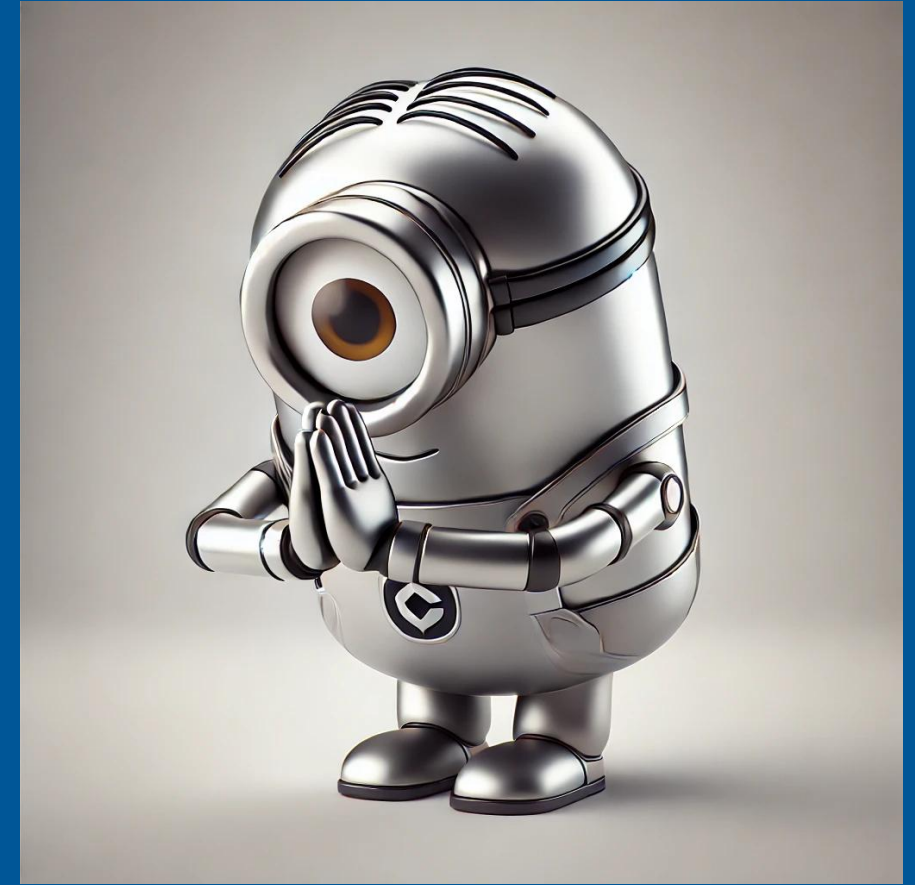
- <https://www.linkedin.com/in/joslat/>

Let me teach you!

- [linkedin.com/learning/instructors/jose-latorre](https://www.linkedin.com/learning/instructors/jose-latorre)

Try maf-Autopilot:

- <https://github.com/joslat/maf-doctor>



Thank You!!

Until next time!!

